

MUHAMMAD MAROOF

Karachi, Pakistan | github.com/maroofiums | linkedin.com/in/maroofiums | kaggle.com/maroofiums | leetcode.com/maroofiums

SUMMARY

Junior ML/AI Engineer with hands-on experience in deep learning, NLP, and production API development. Authored independent research on learnable attention normalization in Transformer architectures (NeurIPS 2026 workshop, under review). Proficient in PyTorch, HuggingFace Transformers, LangChain, FastAPI, and MLOps tooling. Focused on bridging research-grade ML with scalable backend systems.

EDUCATION

Memon Industrial and Technical Institute (MITI)

Karachi, Pakistan

Diploma of Associate Engineering — Software Engineering

Expected Oct 2026

- Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Computer Networks, Software Engineering Principles, Operating Systems, Discrete Mathematics.
- Independent Study:** Deep Learning (PyTorch), Transformer Architectures, Reinforcement Learning, Generative AI, MLOps, System Design for ML, and DSA patterns for technical interviews.

EXPERIENCE

Developer Hub Corporation

Remote

ML/AI Engineer Intern

2024

- Designed and fine-tuned NLP pipelines using HuggingFace Transformers for text classification and information extraction.
- Built and deployed REST APIs with FastAPI, containerized via Docker, and integrated into CI/CD pipelines using GitHub Actions.
- Implemented MLflow experiment tracking for hyperparameter tuning and model versioning across training runs.

Elevvo PathWays

Remote

ML API Intern / ML Specialist (Contract)

2024

- Developed ML-powered REST APIs integrating LangChain with HuggingFace and Mistral LLM backends.
- Built a RAG (Retrieval-Augmented Generation) pipeline for document question-answering using vector stores and embedding models.
- Served fine-tuned models via FastAPI endpoints with low-latency inference optimized for production workloads.

RESEARCH

Adaptive Activation Attention (A³) | Independent Research — Under Review

April 2026

- Proposed replacing fixed softmax in Transformer self-attention with a learnable, input-conditioned gated transformation that adapts attention distributions per input context.
- Evaluated with bootstrap confidence intervals and explicit p-value reporting; includes null result analysis for scientific transparency. Submitted to NeurIPS 2026 workshop (Empirical Findings track).
- Links:** Zenodo DOI | Kaggle Notebook | GitHub

PROJECTS

CognifyAI — AI Code Analysis Platform | React, FastAPI, LangChain, HuggingFace | GitHub

2026

- Built a full-stack AI-assisted code analysis platform with a Monaco Editor frontend and a multi-stage FastAPI backend pipeline: syntax validation → bug detection → security scanning → complexity analysis → optimization → docstring generation.
- Combined AST-based static analysis with LLM-powered enrichment via LangChain; outputs line-level bug reports, CVE-mapped security findings, time/space complexity estimates, and an overall quality score.

Fake News Detection | RoBERTa, FastAPI, React (Vite + TypeScript) | GitHub

2025

- Developed an end-to-end fake news detection web app: fine-tuned a RoBERTa model on a labeled news corpus for binary classification (REAL / FAKE) with confidence scoring.
- Served the model via a FastAPI inference endpoint and built a React (Vite + TypeScript) frontend for real-time headline/article prediction with probability output.

SmartRec — Hybrid Recommendation System | PyTorch, FastAPI, PostgreSQL | GitHub

2026

- Designed a hybrid recommender combining collaborative filtering and content-based signals via a neural matrix factorization model trained in PyTorch on user-item interaction data.
- Exposed real-time recommendation endpoints through FastAPI with PostgreSQL-backed user and item storage; implemented embedding caching to reduce inference latency.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript (React), Bash

ML / DL: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, LangChain, OpenCV

Architectures: Transformers, BERT, GPT, Encoder-Decoder, CNN, LSTM, RAG, Reinforcement Learning

Backend: FastAPI, Django REST Framework, PostgreSQL, REST API design

MLOps / DevOps: MLflow, Docker, GitHub Actions (CI/CD), AWS (S3, EC2)

Tools: Git, LaTeX, Jupyter, Linux, Kaggle, arXiv